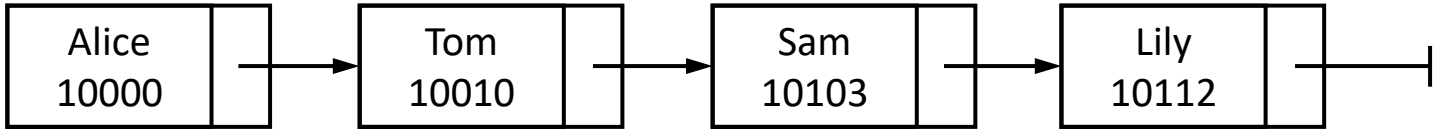
Practice 3. Linked Lists

[CSE2010] Data Structures

Department of Data Science

Overview

- Implement a program for course registration.
- Maintain a linked list of students (with their student IDs and names) who registered a course.
- **TODO**
  - **Add** a student, given a student ID and name.
  - **Delete** the student with a given student ID
  - **Find** a given student ID
- **Constraint**
  - **The list MUST be always sorted in the ascending order of student ID.**



Input File

- Input file contains a list of following operations.
- Each line represents a single operation.
  - **A<space>[ID]<space>[name]**
    - If [ID] already exists in a list, write a failure message to output file.
    - Otherwise, add the student, and write the updated list (to which the student is added) in the output file.
  - **D<space>[ID]**
    - If [ID] does not exist in the list, write a failure message to output file.
    - Otherwise, delete the student and write the updated list (from which the student is deleted) to output file.
  - **F<space>[ID]**
    - If [ID] does not exist in the list, write a failure message to the output file.
    - Otherwise, write the student ID and name of that student to output file.

Input and Output Examples

- Input file

```
A 10003 Mike
A 10000 Alice
A 10010 Tom
A 10010 Tim
F 10001
F 10000
A 10103 Sam
D 10003
D 10003
A 10112 Lily
```

- Output file

```
10003 Mike
10000 Alice 10003 Mike
10000 Alice 10003 Mike 10010 Tom
Addition failed
Search failed
10000 Alice
10000 Alice 10003 Mike 10010 Tom 10103 Sam
10000 Alice 10010 Tom 10103 Sam
Deletion failed
10000 Alice 10010 Tom 10103 Sam 10112 Lily
```

Running Example

- Refer to the setup practice (practice 1) on file I/O to implement this exercise.
- You can implement your code based on the provided **skeleton code**.

```
[hjkim@localhost linked_lists]$ make all
g++ -std=c++11 -o course course.cpp
[hjkim@localhost linked_lists]$ make run
./course input.txt output.txt
[hjkim@localhost linked_lists]$ cat input.txt
A 10003 Mike
A 10000 Alice
A 10010 Tom
A 10010 Tim
F 10001
F 10000
A 10103 Sam
D 10003
D 10003
A 10112 Lily
[hjkim@localhost linked_lists]$ cat output.txt
10003 Mike
10000 Alice 10003 Mike
10000 Alice 10003 Mike 10010 Tom
Addition failed
Search failed
10000 Alice
10000 Alice 10003 Mike 10010 Tom 10103 Sam
10000 Alice 10010 Tom 10103 Sam
Deletion failed
10000 Alice 10010 Tom 10103 Sam 10112 Lily
```

File I/O in Python

- Opening and closing a file
  - f = open(pathToFile, mode)
  - f.close();
- Instead, use **context manager**
  - with open(pathToFile, mode) as f:
- Reading all the lines in a file
  - lines = f.readlines()
- Writing to a file
  - f.write()
- Execution
  - python io.py input.txt output.txt

```
import sys

def execute(inFile, outFile):
  lines = inFile.readlines()
  for line in lines:
    words = line.split()
    cap = float(words[1]) * float(words[2])
    outFile.write(words[0] + " " + str(cap) + "\n")

if __name__ == "__main__":
  if len(sys.argv) != 3:
    print("[exe] [input_file] [output_file]")
    sys.exit(1)
  with open(sys.argv[1], "r") as i, open(sys.argv[2], "w") as o:
    execute(i, o)
```

File I/O in C++

- Opening a file
  - ifstream file; //or ofstream file;
  - file.open(pathToFile);
- Closing a file
  - file.close();
- Reading a file
  - getline(file, string);
- Writing to a file
  - file <<
- Compilation and execution
  - g++ -std=c++11 –o cio io.cpp
  - ./cio input.txt output.txt

```
#include <cstdlib>
#include <iostream>
#include <fstream>
#include <sstream>
using namespace std;
int main(int argc, char* argv[]){
  if (argc != 3) {
    cerr<<"[exe] [input_file] [output_file]"<<endl;
    exit(1);
  }
  ifstream inFile(argv[1]);
  ofstream outFile(argv[2]);
  string line, ticker;
  double price, cap;
  unsigned long long nShares;
  while(getline(inFile, line)) {
    istringstream iss(line);
    iss >> ticker >> price >> nShares;
    cap = price * nShares;
    outFile << ticker << " " << fixed << cap << endl;
  }
  outFile.close();
  inFile.close();
}
```

File I/O in C

- Opening a file
  - FILE* fp = fopen(pathToFile, mode);
- Closing a file
  - fclose(fp);
- Reading a file
  - fscanf(fp, format, variables);
  - (alternatively, use function getline)
- Writing to a file
  - fprintf(fp, format, variables)
- Compilation and execution
  - gcc –o cppio io.c
  - ./cppio input.txt output.txt

```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char* argv[]) {
  if (argc != 3) {
    printf("[exe] [input_file] [output_file]\n");
    exit(1);
  }
  FILE* inPtr = fopen(argv[1], "r");
  if (inPtr == NULL) {
    printf("no such file %s\n", argv[1]);
    exit(1);
  }
  FILE* outPtr = fopen(argv[2], "w");
  if (outPtr == NULL) {
    printf("no such file %s\n", argv[2]);
    exit(1);
  }
  char ticker[256];
  double price, cap;
  unsigned long long nShares;
  while (fscanf(inPtr, "%s %lf %llu", ticker, &price, &nShares) == 3) {
    cap = price * nShares;
    fprintf(outPtr, "%s %f\n", ticker, cap);
  }
  fclose(outPtr);
  fclose(inPtr);
}
```

Submission Guideline

- Submit a compressed (zip) file that contains the following:
  - **source code**
  - **your own input and output files**
- Where: Assignments in LMS
- Deadline: **23:59, March. 24th (Sunday)**